



PINEAPPLE



Visvesvaraya Trade Promotion Centre (VTPC)

VTPC Building, Kasturba Road, Bengaluru-560001

Preface

With a robust agro potential the Country has, the Government of India has launched the One District One Focus Product Scheme (ODOFP) for agriculture sector by the Ministry of Food Processing Industries. Among its primary objectives, the ODOFP initiative aims to enhance the value of the products which is eventually expected to boost employment and income levels of the farmers. The identified products across the country is available at <https://www.nfsm.gov.in/odopstatecropsreport.aspx>

Karnataka has identified its product mix with high demand and export potential across all the districts. The implementing agency for this initiative is the Karnataka State Agricultural Produce Processing and Export Corporation Limited (KAPPEC), the nodal agency in the State.

VTTC being the nodal agency for promotion of exports from the State has made an attempt to draw the action plan district wise, to capture the potential, present status and future prospects in domestic and international markets. Besides, detailed insights have been provided into the biological description of the product, their local, national and international varieties, export-import analysis, HS codes. For a holistic perspective for those concerned, each report also provides SPS standards, processing technologies available, export grading and packing specifications, and suggested pivotal roles and responsibilities among the government departments, boards, corporations and Universities.

Presenting the Action Plan/Report for Pineapple, an ODOFP product mapped to Shivamogga district of Karnataka, formulated by VTTC. I sincerely hope that this ready reckoner with first-hand information regarding the farming sector would help all those concerned, especially the FPOs and food processing entrepreneurs who have a desire to focus on exporting this product.

S.R Satheesha

Director (Exports) & Managing Director

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Introduction

Pineapple (*Ananas comosus*) is a tropical plant with an edible fruit; it is the most economically significant plant in the family Bromeliaceae. The fruit is indigenous to South America, where it has been cultivated for many centuries. Pineapples grow as a small shrub; the individual flowers of the unpollinated plant fuse to form a multiple fruit. The plant is normally propagated from the offset produced at the top of the fruit, or from a side shoot, and typically matures within a year.

Pineapple is an herbaceous perennial, which grows to 1.0 to 1.5 m (3 ft 3 in to 4 ft 11 in) tall. It has 30 or more narrow, fleshy, trough-shaped leaves that are 30 to 100 cm (1 to 3+1/2 ft) long, surrounding a thick stem. The leaves have sharp spines along the margins. The plant produces up to 200 flowers during the season, although some large-fruited cultivars can exceed this. The ovaries develop into berries, which coalesce into a large, compact, multiple fruit. After the first fruit is produced, side shoots (called 'suckers' by commercial growers) are produced in the leaf axils of the main stem. These suckers may be removed for propagation or left to produce additional fruits on the original plant. Commercially, suckers that appear around the base are cultivated.

Pineapple grows in almost any type of soil, provided it is free draining. Slightly acidic soil with pH range of 5.5 to 6.0 is considered optimum for pineapple cultivation. The soil should be well drained and light in texture. Heavy clay soil is not suitable. Sandy loam soils are ideal for pineapple cultivation.

For more information on cultivation -

https://kvk.icar.gov.in/API/Content/PPupload/k0211_47.pdf

Nutritive Value of Pineapple:

Pineapple is a good source of vitamins, minerals like calcium, magnesium, iron etc. and fiber. Pineapples are packed full of vitamin C that boosts our immune system. They contain the enzyme 'Bromelain' in their fibers which is considered to aid in digestion. Besides the above-mentioned benefits, pineapple is said to be helpful in the treatment of Dyspepsia, Bronchitis, High blood pressure and Arthritis.

Varieties/ cultivars and their distribution

- *Ananas comosus* var. *bracteatus* - Brazil, Bolivia, Argentina, Paraguay, Ecuador.
- *Ananas comosus* var. *comosus* - Brazil and Paraguay; naturalized in parts of Asia, Africa, Australia, Mexico, Central America, the West Indies, Northern, South America, and various islands in the Pacific.
- *Ananas comosus* var. *erectifolius* - Peru, Ecuador, Colombia, Venezuela, Northern Brazil, French Guiana
- *Ananas comosus* var. *microstachys* from Costa Rica to Paraguay
- *Ananas comosus* var. *parguazensis* - Colombia, Venezuela, Northern Brazil, Guyana, French Guiana.

For more global varieties, visit: <https://nhb.gov.in/pdf/fruits/pineapple/pin013.pdf>

Popular Pineapple varieties across the world

There are more than 37 types of pineapples grown across the world, each with its own unique set of characteristics. Here's a little bit about each of the ten most popular types of pineapples.

Abacaxi Pineapples

Abacaxi pineapples are very sweet and quite resistant to disease, making them a very popular crop to grow. They weigh anywhere from two to eleven pounds. They're spiny and tall, leaves are bluish-green colored with red-purple tinge and their flesh is nearly translucent in color. They're delicious, but too delicate for commercial purposes. It is a well-known variety in the Bahamas, Florida, and Brazil.



Queen Pineapples

Also known as Common Rough Pineapples found mostly in Australia and South Africa. They produce fragrant dark yellow fruits with a small core. Plants are tolerant to diseases and cold temperatures than other types of pineapple. However, fruits can be eaten raw or fresh as shelf life of canned fruits is less. The leading cultivators of this variety are the Philippines, Queensland, and South Africa.

Red Spanish Pineapples

Aptly named, Red Spanish pineapples are orange red in color and are grown in the Caribbean. It produces fibrous fruit that has a light-yellow color and an aromatic flavor. They're sturdy and high in fiber hence they can be shipped well and are transported all over the world. It is produced by West Indies, Mexico, and Venezuela.



Brecheche

Brecheche pineapples are small, cylinder-shaped, and olive colored. The plant is completely spike-free. The flesh of Brecheche is yellow, very fragrant, and juicy with a small core.



Smooth Cayenne Pineapples

Found in typical food stores and supermarkets in the United States, these fruits have yellow flesh and orange rind with high acidic juice content. Since its flesh has low fiber, it comprises of about 90% of the canned fruit in the world. They're the primary type of pineapple grown in Hawaii, and are bigger than most other varieties, weighing at 4-10 pounds.

Pernambuco Pineapples

Pernambuco pineapples are mostly grown in Brazil. They're white, tender, and juicy with a mild flavor. They're yellow both inside and out and feature distinctive long, spiny leaves. Their cultivation is restricted to South American countries like Ecuador, Colombia, Brazil, Peru, and Venezuela.



Mordilona Pineapples

Mordilona pineapples are grown in the northeastern Andes of Colombia and Venezuela. They're large, irregular, and cylindrical with a yellow to orange peel. The flesh is cream-colored and sweet.

Kona Sugarloaf

Kona Sugarloaf is a type of Smooth Cayenne pineapple. It features white flesh with a high sugar content and no acidity. It's unusually sweet and delicious. One distinctive feature of this type of pineapple is tender and completely edible core instead of usual woody ones.



Singapore Red

Singapore Red pineapples feature green leaves with a reddish stripe at the end. The fruits are reddish, cylindrical, and small with golden-yellow flesh. They're disease and pest-resistant and are great for canning.



Panare

This pineapple is named after the Indian tribe who grew it in Venezuela and is "typical" Pineapple found at the local grocery store or farmer's market. The fruit is bottle-shaped and orange-colored. The flesh is deep yellow, slightly fragrant with slightly sweet flavor and creamy texture. It has a small core. They have various stages of ripening, from green to yellow, with brown spots. They're grown across Central America, and their production is essential to the economies in these areas.



Roja Espanola Pineapple

These are barrel shaped and have yellow-orange colored flesh. They consist of small single fruits that gather around to make a pineapple. The taste is pleasant and sweet.

Some other common types of pineapples are Ripley, Natal Queen, Green Spanish, Roja Espanola, James Queen, Baron de Rothschild, Montufar, Delmonte Gold, White Kauai, MD2, Burmanguese, Antigua Black, Queen, Monte Oscuro, Champaka, MacGregor, Red Spanish, Cabezona, PR 1-67, Valera, St. Michael, Giant Kew, Masmerah, Hilo, Mauritius Pernambuco, Singapore Spanish, Rondon, Perolera, Spanish Jewel, Pinkglow, White Jade, Maui Gold, Monte Lirio, Maipure and Mordilona.



Source:

<https://www.crazymasalafood.com/top-20-different-types-of-pineapple-you-should-know/>

Varieties of commercially grown Pineapples in India

Most common varieties grown in India are - Kew, Giant Kew, Queen, Mauritius, Jaldhup, Lakhat etc. Among these varieties, Kew/ Giant Kew and Queen are commonly grown in the northeastern part of India and is believed to be best suited for these parts of the country.

Varieties	States
Kew, Mauritius, Queen	Assam and other Northeast states
Queen, Giant Kew	West Bengal
Mauritius, Queen, Kew	Kerala
Lakhat, Jaldhup	Assam

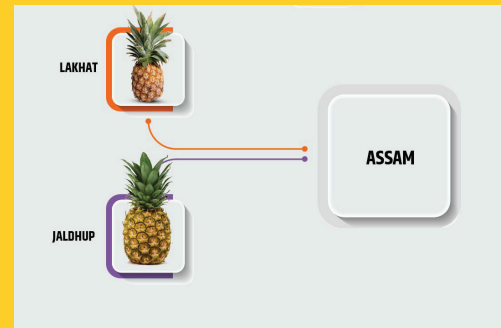
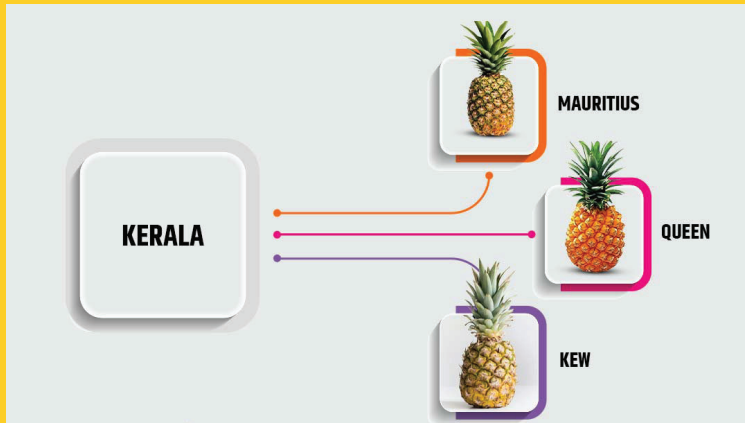
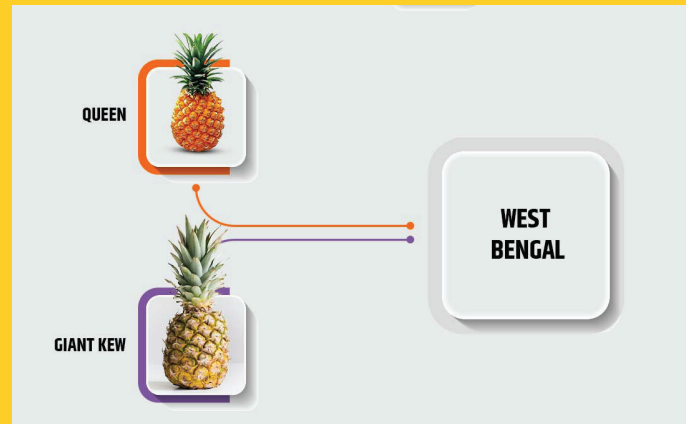
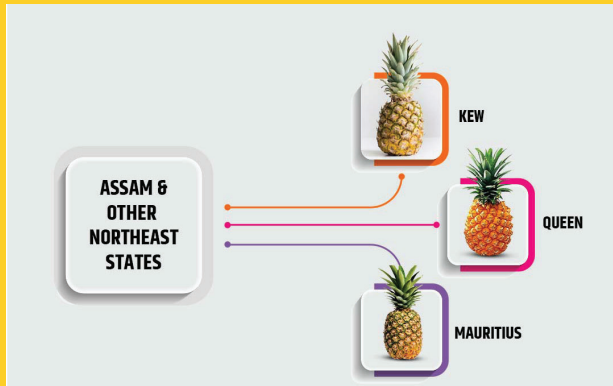
States	Cultivation Time
Kerala and Karnataka	April to June
Assam and other North-eastern States	August to October
West Bengal Northern parts	October to November
West Bengal Other Parts	June to July

Source:

<https://www.tractorjunction.com/blog/pineapple-farming-varieties-and-health-benefits/>



Pineapples Varieties in Different States



Production data of Pineapples (World – 2020)

Rank	Country	% Share	Production
1	Phillipines	9.73%	2.70M
2	Costa Rica	9.45%	2.62M
3	Brazil	8.84%	2.46M
4	Indonesia	8.81%	2.45M
5	China	8.00%	2.22M
6	India	6.48%	1.80M
7	Thailand	5.52%	1.53M
8	Nigeria	5.43%	1.51M
9	Mexico	4.35%	1.21M
10	Columbia	3.18%	882.63K

Source:

<https://www.tridge.com/intelligences/pineapple/production>

State-wise production data of Pineapples (2021-22 Est):

No.	States/UTs	Area	Production
1	West Bengal	11.95	356.32
2	Assam	17.85	338.98
3	Karnataka	2.96	169.54
4	Tripura	9.92	144.6
5	Meghalaya	12.22	138.93
6	Manipur	12.12	134.82
7	Nagaland	8.93	114.77
8	Bihar	3.9	113.76
9	Kerala	9.81	89.05
10	Andhra Pradesh	4.17	66.77
	TOTAL	105.85	1807.62

Source:

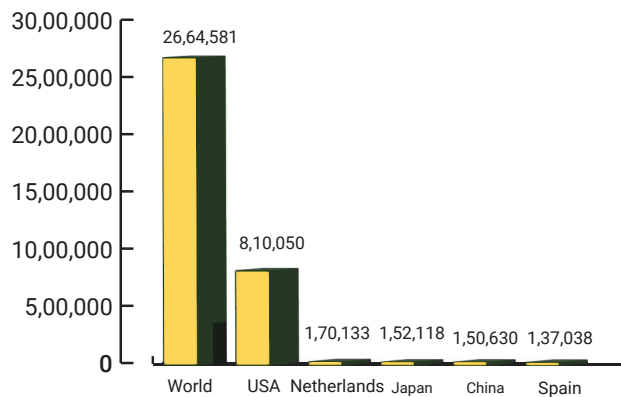
https://agricoop.nic.in/sites/default/files/2021-22%20%28First%20Advance%20Estimates%29%20%281%29_0.pdf

World Export and Imports

080430: Fresh or dried pineapples

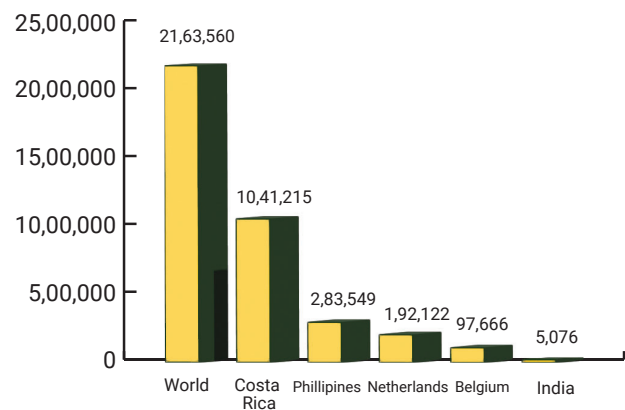
2021 Global Imports 2,664,581 USD.

No.	Country	Value USD
1	USA	810,050
2	Netherlands	170,133
3	Japan	152,118
4	China	150,630
5	Spain	137,038



2021 Global Exports 2,163,560 USD.

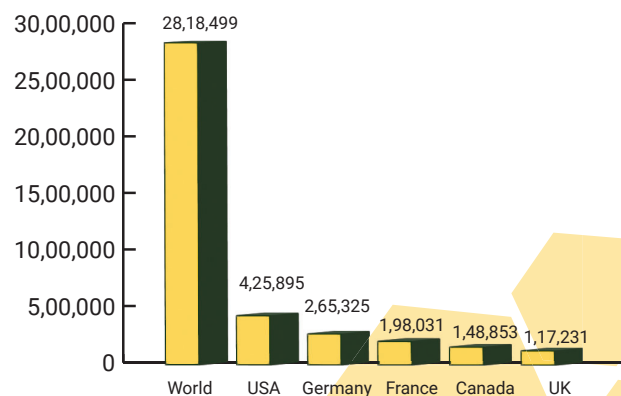
No.	Country	Value USD
1	Costa Rica	1,041,215
2	Phillipines	283,549
3	Netherlands	192,122
4	Belgium	97,666
27	India	5,076



200799: Jams, jellies, marmalades, purées or pastes of fruit, obtained by cooking, whether or not containing added sugar or other sweetening matter (excluding citrus fruit and homogenized preparations of subheading 2007.10)

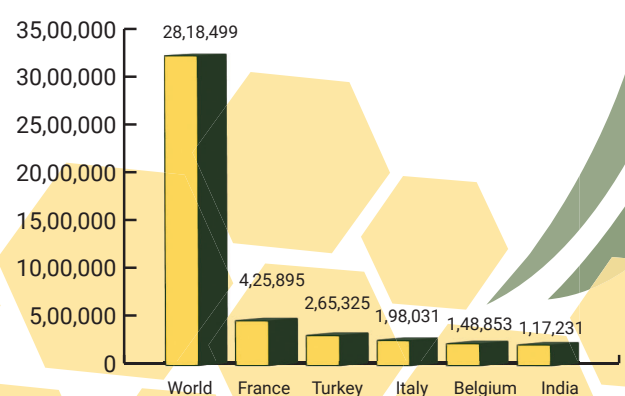
2021 Global Imports 2,818,499 USD.

No.	Country	Value USD
1	USA	425,895
2	Germany	265,325
3	France	198,031
4	Canada	148,853
5	United Kingdom	117,231



2021 Global Exports 3,212,016 USD.

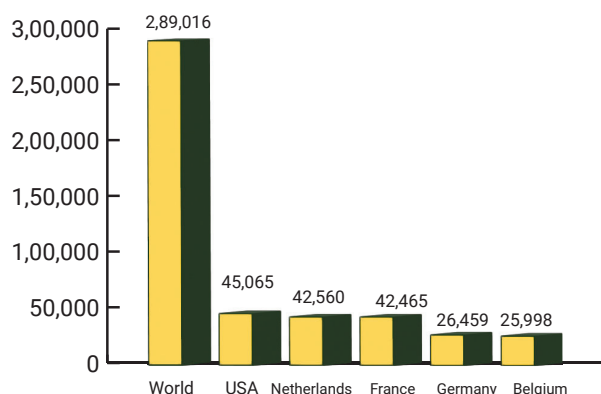
No.	Country	Value USD
1	France	448,652
2	Turkey	297,459
3	Italy	247,227
4	Belgium	213,880
6	India	200,983



200941: Pineapple juice, unfermented, Brix value ≤ 20 at 20°C, whether or not containing added sugar or other sweetening matter (excluding containing spirit)

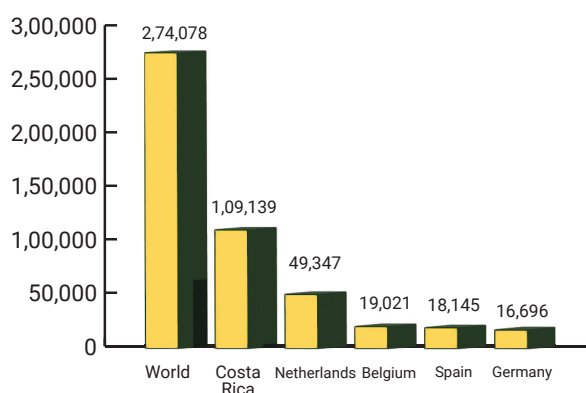
2021 Global Imports 289,016 USD.

No.	Country	Value USD
1	USA	45,065
2	Netherlands	42,560
3	France	42,465
4	Germany	26,459
5	Belgium	25,998



2021 Global Exports 274,078 USD.

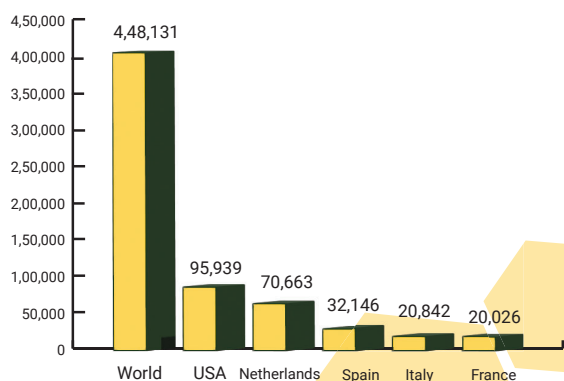
No.	Country	Value USD
1	Costa Rica	109,139
2	Netherlands	49,347
3	Belgium	19,021
4	Spain	18,145
5	Germany	16,696



200949: Pineapple juice, unfermented, Brix value > 20 at 20°C, whether or not containing added sugar or other sweetening matter (excluding containing spirit)

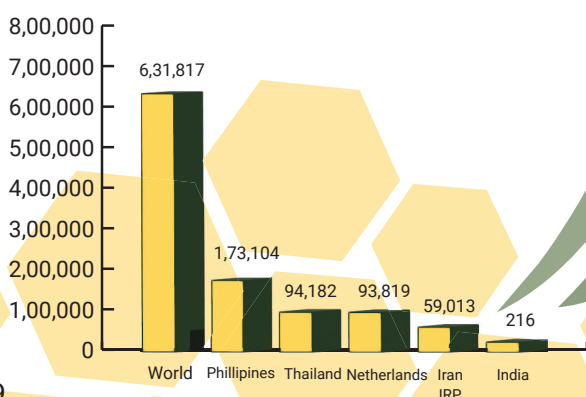
2021 Global Imports 448,131 USD.

No.	Country	Value USD
1	USA	95,939
2	Netherlands	70,663
3	Spain	32,146
4	Italy	20,842
5	France	20,026



2021 Global Exports 631,817 USD.

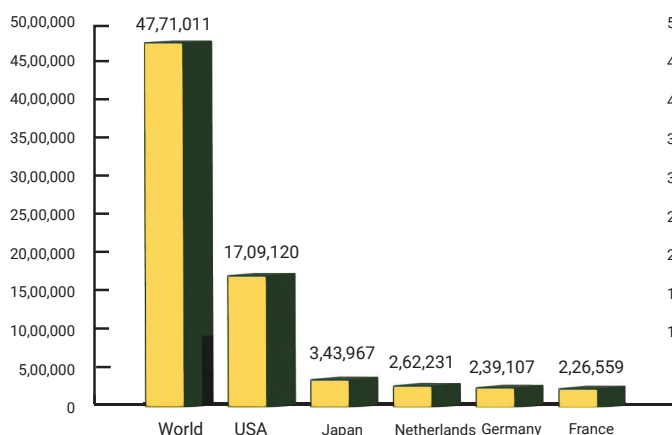
No.	Country	Value USD
1	Phillipines	173,104
2	Thailand	94,182
3	Netherlands	93,819
4	Iran IRP	59,013
40	India	216



200899: Fruit and other edible parts of plants, prepared or preserved, whether or not containing added sugar or other sweetening matter or spirit (excluding prepared or preserved with vinegar, preserved with sugar but not laid in syrup, jams, fruit jellies, marmalades, fruit purée and pastes, obtained by cooking, and nuts, groundnuts and other seeds, pineapples, citrus fruits, pears, apricots, cherries, peaches, strawberries, palm hearts and cranberries)

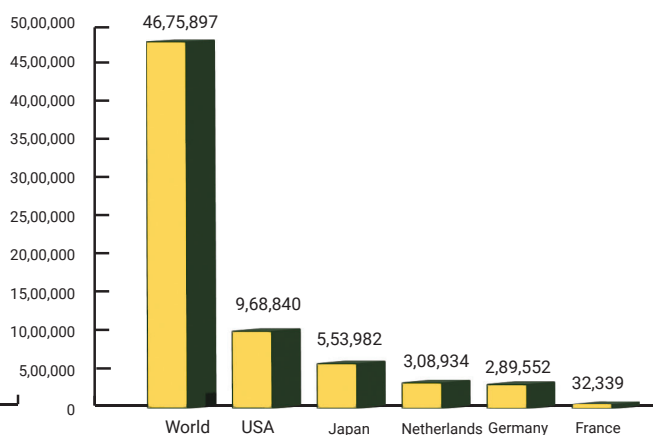
2021 Global Imports 4,771,011 USD.

No.	Country	Value USD
1	USA	1,709,120
2	Japan	343,967
3	Netherlands	262,631
4	Germany	239,107
5	France	226,559



2021 Global Exports 4,675,897 USD.

No.	Country	Value USD
1	China	968,840
2	Korea RP	553,982
3	Mexico	308,934
4	Netherlands	289,552
27	India	32,339

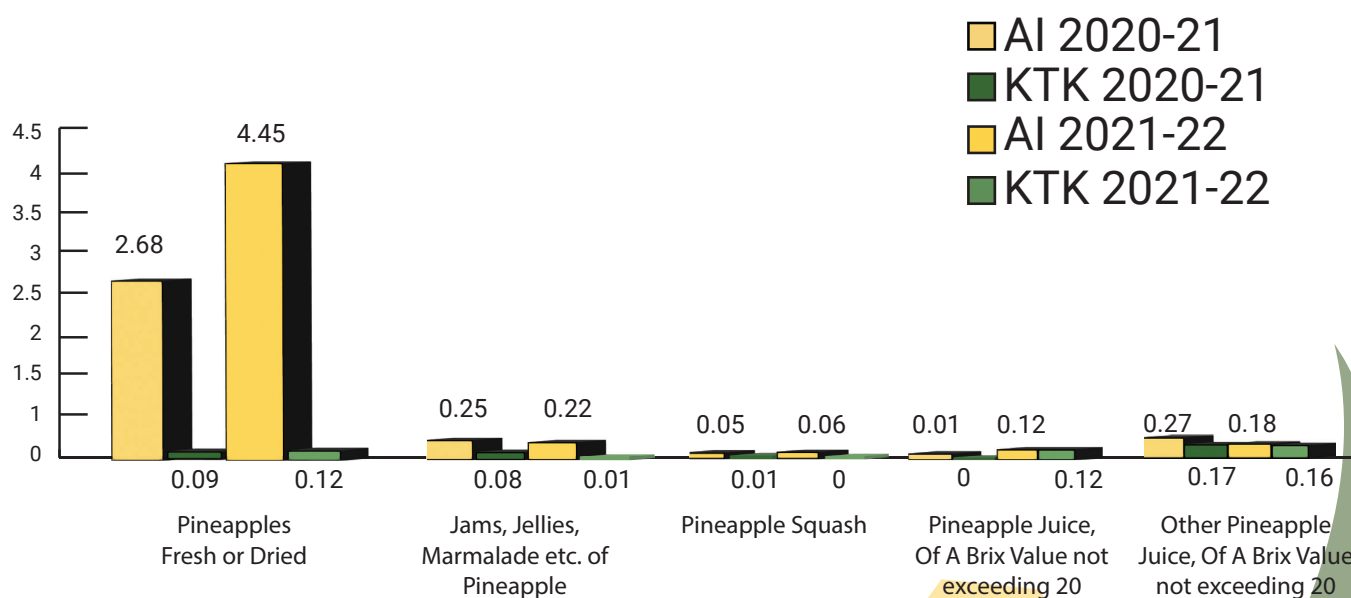


Source: ITC Trade map. Year - 2021.



Export performance of Pineapple in the last two years: All India and Karnataka (Value in Mn USD)

No.	HS Code	Description	2020-21		2021-22		Major Importing Countries
			AI	KTK	AI	KTK	
1	08043000	Pineapples Fresh or Dried	2.68	0.09	4.45	0.12	UAE, Nepal and Qatar
2	20079930	Jams, Jellies, Marmalade etc. of Pineapple	0.25	0.08	0.22	0.01	UAE, USA and Nepal
3	20089914	Pineapple Squash	0.05	0.01	0.06	Nil	USA, Nepal and Qatar
4	20094100	Pineapple Juice, Of A Brix Value not exceeding 20	0.01	Nil	0.12	0.12	USA, Netherland and UAE
5	20094900	Other Pineapple Juice, Of A Brix Value not exceeding 20	0.27	0.17	0.18	0.16	Germany, Netherland and Nepal

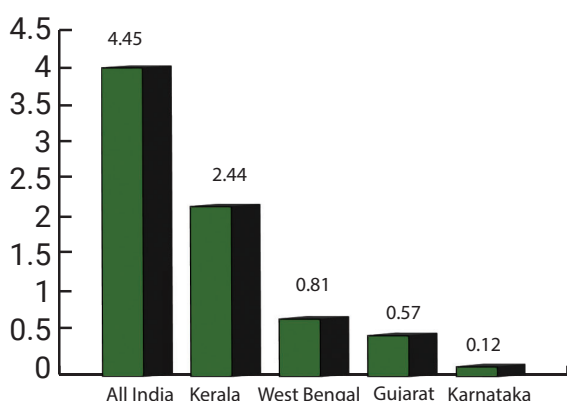


Export Analysis of Pineapples: 2021-22

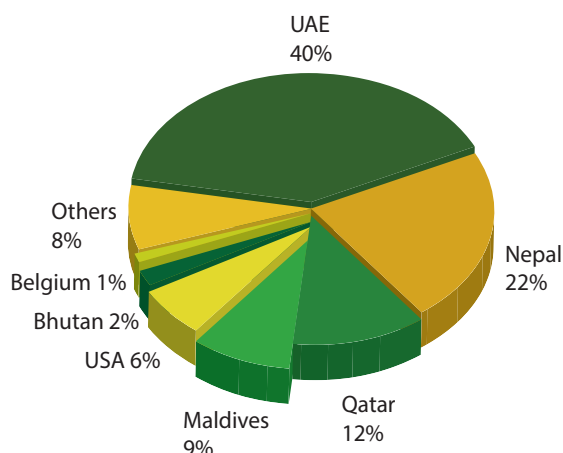
08043000: Pineapples Fresh or Dried

No.	State	Value USD Mn	Destination
	All India	4.45	UAE, Nepal, Qatar (+19)
1	Kerala	2.44	UAE, Qatar, Maldives(+9)
2	West Bengal	0.81	Nepal & Bhutan
3	Gujarat	0.57	USA, Qatar, Canada, Aruba
6	Karnataka	0.12	Nepal, Qatar, Maldives, Saudi Arabia

Exporting States



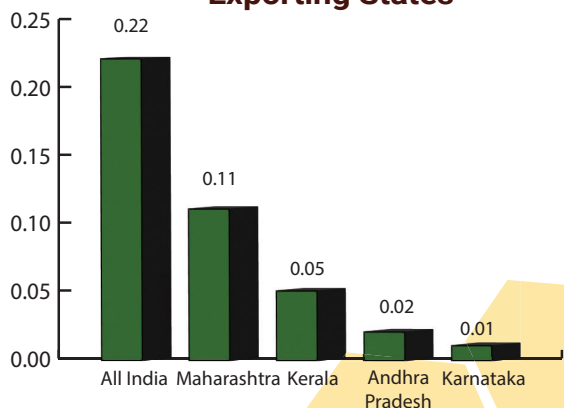
Importing Countries



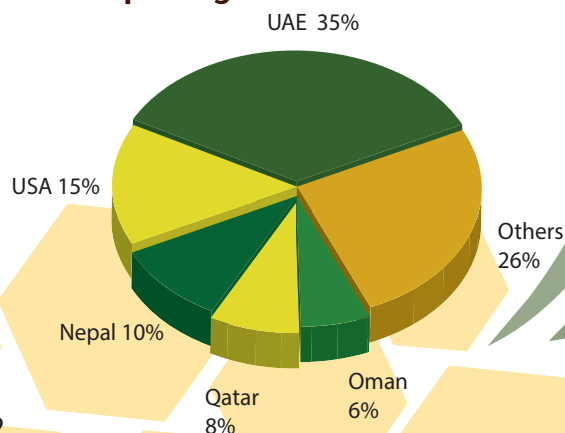
20079930: Jams, Jellies, Marmalade etc. of Pineapple

No.	State	Value USD Mn	Destination
	All India	0.22	UAE, USA & Nepal (+26)
1	Maharashtra	0.11	UAE, Qatar, Oman (+10)
2	Kerala	0.05	UAE, Kuwait, Saudi Arabia (+14)
3	Andhra Pradesh	0.02	USA
6	Karnataka	0.01	Nepal, Qatar, Maldives, Saudi Arabia

Exporting States



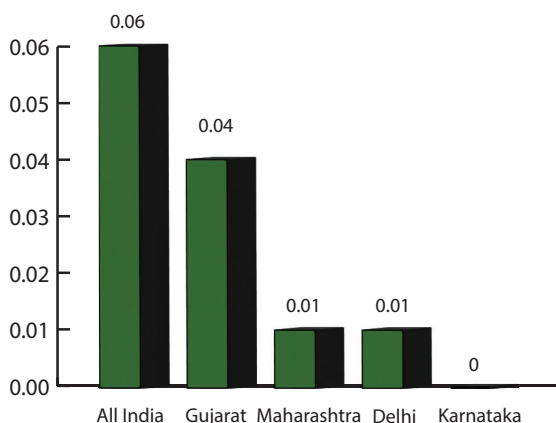
Importing Countries



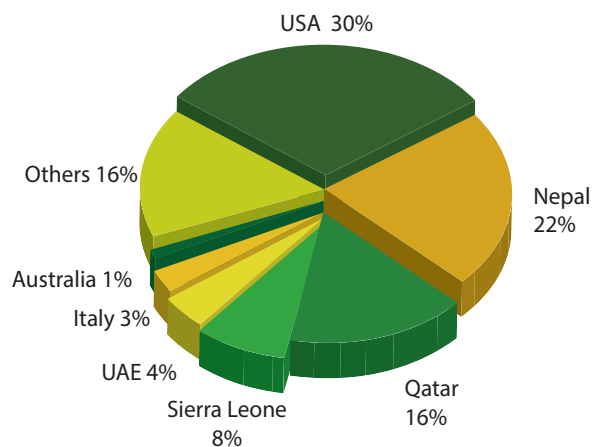
20089914: Pineapple Squash

No.	State	Value USD Mn	Destination
	All India	0.06	USA, Nepal, Qatar (+21)
1	Gujarat	0.04	USA, Sierra Leone, Qatar (+3)
2	Maharashtra	0.01	Nepal, USA, Qatar (+6)
3	Delhi	0.01	Italy, Australia, Hong Kong (+3)
0	Karnataka	Nil	Nil

Exporting States



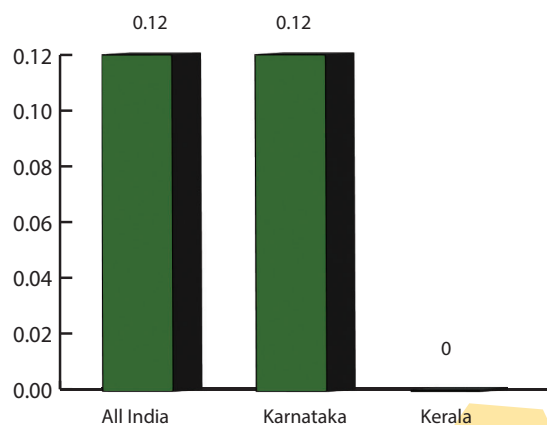
Importing Countries



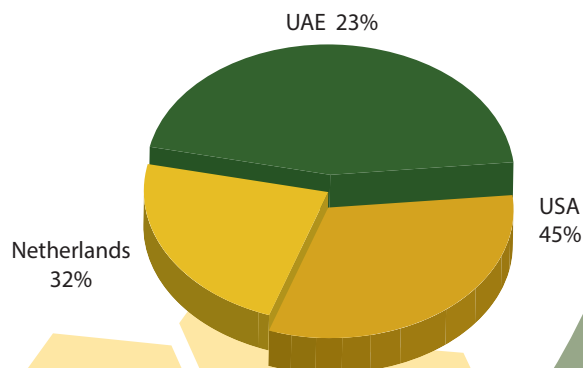
20094100: Pineapple Juice, Of A Brix Value not exceeding 20

No.	State	Value USD Mn	Destination
	All India	0.12	USA, Netherlands, UAE
1	Karnataka	0.12	USA & Netherlands
2	Kerala	0	UAE

Exporting States



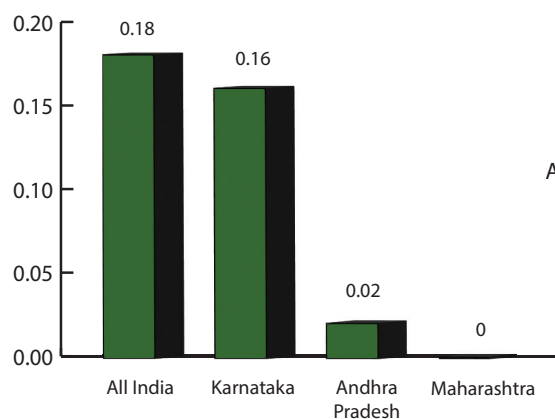
Importing Countries



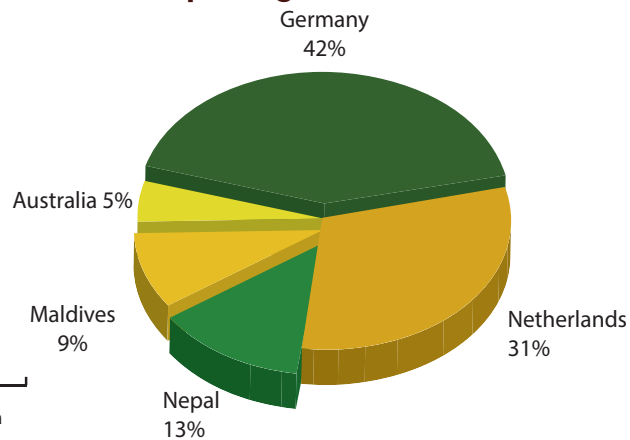
20094900: Other Pineapple Juice, Of A Brix Value not exceeding 20

No.	State	Value USD Mn	Destination
	All India	0.18	Germany, Netherlands, Nepal (+2)
1	Karnataka	0.16	Germany & Netherlands
2	Andhra Pradesh	0.02	Germany
3	Maharashtra	0.00	Nepal & Maldives

Exporting States



Importing Countries



Source:

Exim Analytics, DGCIS, 2021-22



SPS -TBT Standards for Pineapple

Pre-harvest practices for selection of export quality fruits:

- Whole fruit with crown
- Crown should be free of dead and dried leaves
- Healthy produce generally affected by rotting or deterioration to make the fruit unfit for consumption
- Fruits should be clean, practically free from any visible foreign matter
- Fruits should be free from internal browning
- Fruits should be free from damage caused by pests
- Free from pronounced blemishes
- Free from damages caused by low/higher temperature in field conditions
- Free from external moisture, excluding condensation following removal from cold storage
- When a peduncle is present, it shall not be longer than 2 cm and the cut must be straight and clean.

Post-harvest steps to be followed for exports:

- Harvesting of 1/8th matured fruit (color) must be done in the early morning or late afternoon (Temp 25-30oC).
- Harvested crop must be stored in plastic crates under shady conditions in the field.
- Freshly packed harvest must be transported to packhouse with the receipt of the produce & record of the pre-harvest management.
- Sorting, grading and selection of good fruits happens at the packhouse.
- A sharp knife will be used to trim/cut the stem at 2.0cm length.
- Selected fruits are subjected to washing with soft potable water followed by hot water treatment at 53oC for 5-7 minutes and then fruits are treated with a fungicide.
- Treated fruits are allowed for air drying.
- Air dried fruits are waxed, packed, labeled, and palletized. These pallets are subjected to pre-cooling at 13-15oC for 6-8 hours and stored in reefer containers at 12-12.5oC and 85% relative humidity. Reefers are transported to exit ports.

For more information visit:

https://kau.in/sites/default/files/documents/protocol_for_export_of_pineapple.pdf

Measures for Enhancing Competitiveness for Exports by APEDA

In a bid to harness the export potential of naturally grown fresh pineapples from the northeastern states, the Agricultural and Processed Food Products Export Development Authority (APEDA) had organized an in-store export promotion event in Dubai, to promote Assam's pineapples, considered the sweetest in the world, among consumers for wider acceptance in the Gulf countries. With a production of 338.98 metric ton in 2020-21, Assam ranks second in pineapple production having a share of 18.75% in total production in India.

The 'in-store promotion show' was organized in association with Dubai's largest supermarket group, Lulu Group, and the organizers said it was part of APEDA's strategy to promote locally produced agricultural products in the international market. The showcased pineapples are procured from Hmarkhawlien village under Lakhipur sub-division in Cachar district, Assam. Consumers were offered to taste the sweetness of the pineapples, one of the most sought-after tropical fruit rich in fiber.

APEDA chairman - M Angamuthu insisted on promoting the processed form of value-added pineapple from the northeastern states to sustain production in the global export market for a longer time. 'We need to focus on promoting pineapple sourced from farmers in processed form in the Gulf countries in collaboration with private entities, including the Lulu Group. It will help farmers in better price realization of their produce,' he said, emphasizing that 10-15 aggregators should be identified for expediting hassle-free export from the northeast.

Before Assam made its debut in pineapple export to Dubai in 2019, Tripura was the first northeastern state to export its 'queen' variety of pineapple to Dubai and Doha in 2018. Tripura's pineapple was also exported to Bangladesh in 2020. APEDA informed that the top 10 importing countries of Indian pineapples are UAE, Nepal, Qatar, Maldives, USA, Bhutan, Belgium, Iran, Bahrain and Oman. 'A quantity of 7665.42 MT of pineapples valued at \$4.45 million was exported in the year 2021-22,' it stated.

APEDA said the northeastern region witnessed an 85.34% growth in the export of agricultural products in the last six years as it increased from \$2.52 million in 2016-17 to \$17.2 million in 2021-22. The major destination of export has been Bangladesh, Bhutan, the Middle East, the UK and Europe.

Action plan

- Production technology on modern lines needs to be demonstrated to the growers on a massive scale.
- Farmers must be encouraged to grow new and improved varieties of pineapples.
- Farmers need to be educated about export requirements and international quality standards of pineapples.
- Protocol for pre- and post-harvest handling of Pineapples need to be perfected/ standardized for shipping to Gulf countries by sea.
- Most modern pack house facilities need to be created, to begin within the pineapple growing regions of Northern part of Karnataka.
- Learning from the North-eastern states of India, groups of farmers need to be encouraged to have a system of makeshift packhouses, to cut and pack in the field itself and send these sorted fruits to a central packhouse for further processing, treatment and packing.
- It is suggested conduct of training to packhouse workers about post-harvest handling technology and impart knowledge about international quality standards.

Opportunities to Explore

- Though India has a substantial position in fresh fruit exports of Pineapple, we have very less exports in value-added products like Jams, Jellies, Marmalades.
- India's exports of pineapple juices is very minimal which paves way and opportunities for many entrepreneurs and processing industries to explore.
- Establishment of such processing industries will focus on stepping up production of pineapples, in turn motivating the farmers and growers to indulge themselves in pineapple farming.



Responsibility Matrix

No.	Actions	Responsibility
1	Research on best global varieties.	Agricultural and Horticultural Universities, Karnataka.
2	Delegation of progressive farmers to Costa Rica, Brazil and Philippines	Horticulture department, Karnataka
3	Farmer Sensitization programs on available opportunities.	Agriculture and Horticulture departments, Karnataka
4	Inviting Exporters/entrepreneurs to Investors meet for contract farming and processing industries	Horticulture Department and DIC Districts
5	Farmer database and aggregation of the produce.	Horticulture department and related FPOs
6	Processing technology and Value Addition Handholding	Horticulture department through CFTRI, Mysore and IIFPT, Thanjavur
7	Export training, orientation and Market Intelligence	VTPC Karnataka

Regulators and Service providers

No.	Organization	Service	Contact
1	APEDA	RCMC, Market survey and assistance.	1st Floor, Beeja Bhavan, Bellary Rd, Hebbal, Bengaluru - 560024. E: apedabl@apeda.gov.in
2	KAPPEC	PMFME Scheme	17, Richmond Rd, Shanthala Nagar, Richmond Town, Bengaluru - 560025. E: kappec1996@gmail.com
3	DGFT	IEC, Customs and ICEGATE	6th floor, Kendriya Sadan, C & E Wing, 17th main, Koramangala 2nd Block, Koramangala, Bengaluru - 560034 E: bangalore-dgft@nic.in
4	Plant Quarantine	Phyto Sanitary certification (SPS)	Hebbal-Boopasandra Road HA Farm Post, Bengaluru - 560024. E: dd-pqfsb-ka@nic.in
5	BIAL Cool Port	Freight and Flight	KIAL Road, Devanahalli, Bengaluru -560300 W: http://www.aisats.in